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HIGHLIGHTS

- Above-normal snowfall reduces net migration in lake-effect snow counties in the Great Lakes region but not in other counties.
- The migration response to snow in lake-effect counties is driven by reductions in in-migration.
- Snow effects are largest for young and middle-aged adults and dissipate within one to two years.

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ABSTRACT

Do short-run weather shocks affect migration? I estimate how snow influences county-level migration in the Great Lakes region since 1970. I isolate responses to snow by comparing its effect on net migration across regions exposed to lake-effect snow (LES)—heavy snow generated downwind of the Great Lakes. Higher-than-average snowfall in lake-effect regions leads to net population loss the following year. This effect is driven by reductions in in-migrants, dissipates after 1–2 years, and is strongest for young and middle-aged populations. Snow has virtually no effect on migration in non-lake-effect regions, suggesting that baseline exposure interacts with anomalous weather events.

1. Introduction

Climate is a key determinant of household location choice (Graves, 1980). Since the middle of the 20th century, Americans have largely moved from colder regions in the Northeast and Midwest to warmer regions on the coasts and in the Sunbelt (Glaeser and Tobio, 2008; Molloy et al., 2011), potentially increasing overall exposure to extreme heat, drought, sea-level rise, hurricanes, and wildfires (Hsiang et al., 2017). When deciding to move, people choose locations based on a bundle of market and nonmarket amenities, including those derived from local climates, and are willing to accept lower wages and pay higher housing costs to live in areas with more desirable climate attributes (Roback, 1982; Cragg and Kahn, 1997; Timmins, 2007; Albouy et al., 2016; Sinha et al., 2018; Fan et al., 2018; Sinha et al., 2021).¹ Because climate

change is altering the distribution of local climate amenities by increasing temperatures, changing precipitation patterns, and increasing the frequency of extreme weather events (IPCC, 2022), there is heightened interest in how weather and climate affect migration (Cattaneo et al., 2019).

Extant research suggests that people move in response to *fast-onset* extreme weather events and natural disasters, such as hurricanes, floods, and wildfires (e.g., Bohra-Mishra et al., 2014; Eyer et al., 2018; Boustan et al., 2020; Deryugina and Molitor, 2020; Mahajan and Yang, 2020; Winkler and Rouleau, 2021; Bakkensen and Ma, 2020; Sheldon and Zhan, 2022; Wrenn, 2024), as well as *slow-onset* climate change, including drought, sea-level rise, and gradual temperature change (e.g.,

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¹ Recent work also suggests that international migrants seek out climates that are familiar to their origins (Obolensky et al., 2024).

Hornbeck, 2012; Hauer, 2017; Baylis et al., 2025).² This body of work generally suggests that short-run weather anomalies do not affect migration.³ However, there are several challenges in estimating migration responses to weather. First, most research focuses on variation in temperature, which does not change much year-to-year, becomes less remarkable with repeated exposure (Moore et al., 2019), is highly spatially correlated, and may be relatively easy to adapt to in wealthy countries (Barreca et al., 2016). Second, local climates are defined by a highly correlated bundle of climate characteristics (Auffhammer et al., 2013), which makes it difficult to isolate the effect of changes in any one characteristic on migration.

In this paper, I overcome these challenges by investigating how short-run changes in winter weather—snow, in particular—affect net migration over the past half-century. I focus on the US Great Lakes region where lake-effect snow (LES) provides an opportunity to isolate annual variation in snow from contemporaneous changes in weather. Lake-effect snow occurs when cold air moves over unfrozen lakes, picks up moisture, and deposits that moisture as snow on the southeastern side of a Great Lake. This process generates remarkable spatial variation and substantial year-to-year variability in distinct regions downwind of a Great Lake, with annual snowfall often varying by more than 100 percent due to idiosyncratic changes in lake ice cover and wind patterns (Notaro et al., 2015; Burnett et al., 2003; Ellis and Johnson, 2004). Empirically, I explore how snow influences county-level migration since 1970 using a novel data set derived from annual population changes, net of births and deaths, as well as migration data inferred from tax returns. I estimate within-county changes in net migration rates in response to annual deviations in snow from a longer-run average, controlling for weather and unobserved county and state-year effects. I isolate the effect of snow by comparing its marginal effect on migration across counties that typically do or do not receive lake-effect snow. I also explore nonlinearities in the effect of snow, temporal dynamics, differences in in- vs. out-migration rates, regional effects, and age heterogeneity to better elucidate the migration-response to snow.

The primary result suggests that positive deviations in annual snow from longer-run averages (i.e., more snow than usual) lead to net population losses in the following year. The effect is smaller, but remains significant, when using a temporally limited sample of tax filers. The migration-response to snow is driven almost entirely by counties that regularly receive lake-effect snow. These counties exhibit a reduction in net migration rates of 1.5–3 people per 1000 for an abnormally snowy winter in the previous year, whereas there is virtually no effect for non-lake-effect counties. The differential effects across regions are surprising since lake-effect regions are likely better adapted to deal with large amounts of snow, which suggests the results may speak to how (would-be) residents update beliefs about the likelihood of future climate based on realizations of weather (Deryugina, 2013). Nonlinear effects imply that any snow above-normal generates similar levels of population loss, suggesting that the total quantity matters less than a “bad winter.” The migration response is driven by reductions in in-migration, where snow serves as a deterrent for moving to these regions rather than pushing existing residents out. The effect dissipates after 1–2 years, which suggests that the effect is driven by short-run weather shocks rather than longer-run changes in climate. Lastly, the effect is strongest for young and middle-aged adults, with small or null effects for older cohorts at or above the typical retirement age.

These findings contribute to our understanding of weather, climate, and migration in several ways. The first contribution is empirical: snow

is correlated with winter temperatures, which can confound the effect of temperature on migration. Cross-sectional analyses of households’ willingness-to-pay for climate amenities (e.g., Cragg and Kahn, 1997; Albouy et al., 2016; Sinha et al., 2018; Fan et al., 2018; Sinha et al., 2021) may be biased if there are unobserved attributes of a location that correlate with long-run differences in snow (e.g., adaptation to local climates, such as plowing infrastructure to remove snow, is a critical unobserved variable that may affect the influence of snow on quality-of-life measures). Estimating the effect of snow on migration across regions with similar climates—aside from differences in snow—allows for separating snow from confounding temperatures. Accounting for the bundle of experienced climate amenities, and understanding how each of those amenities affects migration decisions, is critical for characterizing how weather affects migration.

Second, snow is a highly salient climate attribute that defines the local climate in regions of the Midwest and Northeast. Additionally, snowfall totals can change by more than 100 percent from year to year, providing substantial variation in regularly experienced weather. Annual changes in temperature are less salient, become less remarkable with repeated exposure (Moore et al., 2019), and are highly spatially correlated (Auffhammer et al., 2013), leading to challenges in identifying their effects, even when using within-county variation in fixed-effects regressions. Further, snow is not typically as damaging as other extreme weather events, such as hurricanes or floods, which can lead to large, one-time migration events. Contrary to previous assertions, this research demonstrates that salient fluctuations in short-run weather do affect migration.

Third, the differential response to snow across regions presents a puzzle. Conventional wisdom suggests that regions that regularly receive large amounts of snow would be better adapted to those conditions. However, the results suggest the opposite: the migration-response to snow is driven almost entirely by lake-effect counties that regularly receive heavy snow. These differential responses may reveal how (potential) residents update their beliefs about future climate (Deryugina, 2013). Abnormally large amounts of snow in lake-effect counties could be indicative of the likelihood of these events occurring in the future. In contrast, residents in non-LES regions may interpret a bad snow year as a one-off event.

Lastly, this research provides several insights for the Great Lakes region—and the Midwest and Northeast, more generally—which have seen their share of the US population decline since the middle of the 20th century. This demographic shift has been attributed to a variety of factors, including the region’s climate (Glaeser, 2007; Molloy et al., 2011; Leduc and Wilson, 2024). The results of this paper suggest that year-to-year changes in snow are a causal factor for population change over this period. The effect is largest for younger adults, which could contribute to “brain drain” in these regions (Gibson and McKenzie, 2011). Despite the relative decline in population over time, Great Lakes cities are often heralded as “climate havens” due to their existing urban infrastructure that can support much larger populations, freshwater access, mild summers, and low climate risks (Kahn, 2013). Knowing whether snow will deter future migrants is important to fully characterize demographic changes and their effects on the regional economy, especially because the region is expected to experience increased lake-effect snowfall through the end of the century (Notaro et al., 2015). Understanding the context of how snow affects migration is important to characterize the future costs of climate change in cold climates.

2. Empirical setting and background on lake-effect snow

To isolate the effect of snow on migration, I restrict my analysis to US counties within 200 km of a Great Lake shoreline, which includes parts of Minnesota, Wisconsin, Illinois, Indiana, Ohio, Pennsylvania, West Virginia, New York, and all of Michigan. This region, often considered the “rust belt,” includes major US cities, such as Chicago, IL; Milwaukee, WI; Detroit, MI; Cleveland, OH; Buffalo, NY; and Rochester,

² See Cattaneo et al. (2019) for a broad review of climate and migration.

³ For example, Albouy et al. (2016) state that annual changes in weather are unlikely to affect a household’s willingness-to-pay to live in a particular area. Baylis et al. (2025), however, find that short-run changes in temperature do affect migration decisions, although the effect is more pronounced when considering longer-run changes.

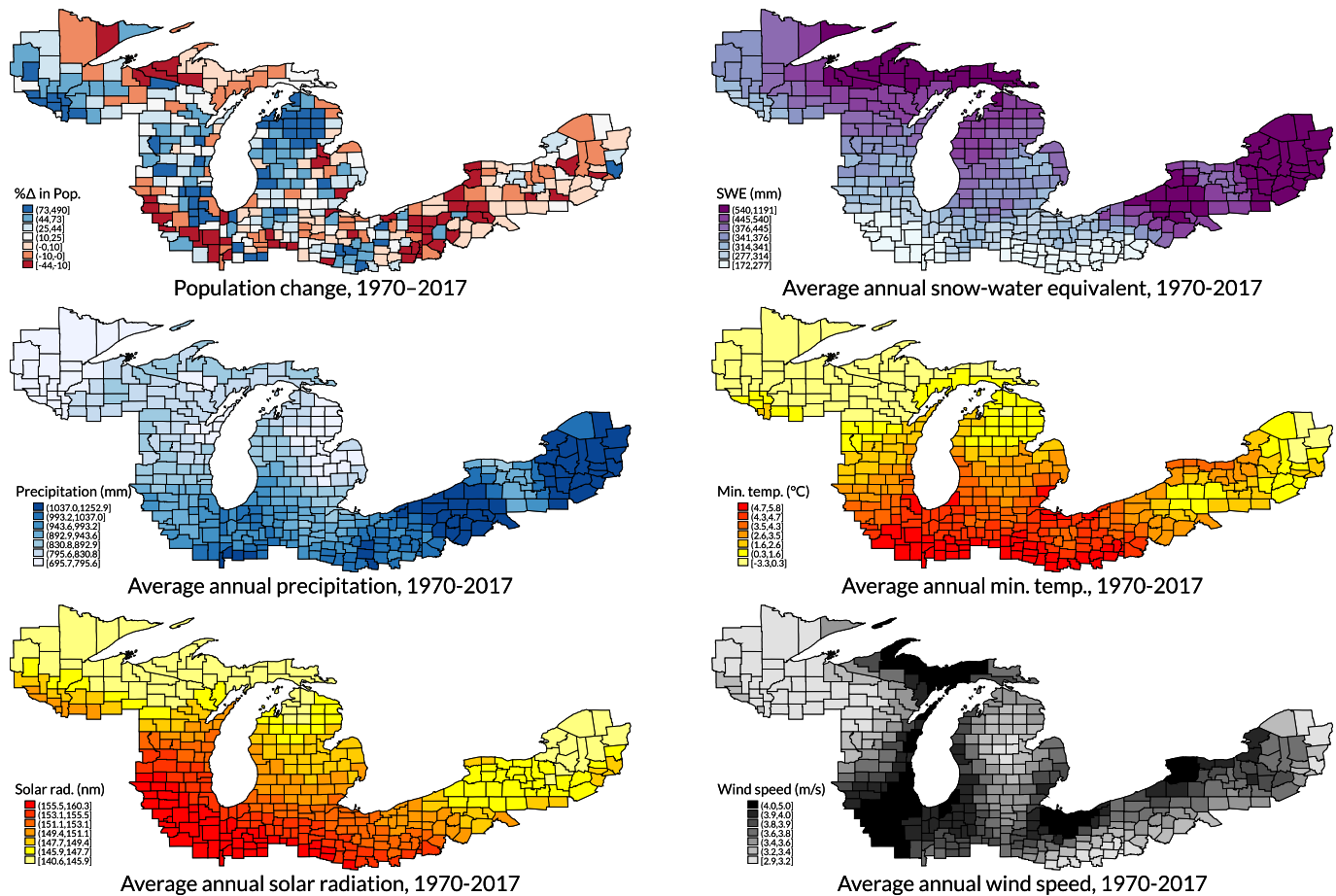


Fig. 1. Maps of population and selected weather variables. *Notes:* Maps display county-level averages over the sample period 1970–2017 for the Great Lakes region, restricted to counties within 200 km of a Great Lakes shoreline. Population change is the percent change in total population between 1970 and 2017. Snow-water equivalent (SWE), precipitation, minimum temperature, solar radiation, and wind speed are annual averages constructed from the TerraClimate dataset (Abatzoglou et al., 2018).

NY. By some measures, the Great Lakes region currently generates more than \$3T in annual economic output, comparable to the GDP of India and larger than that of the UK, France, and Russia.⁴

Although climate is generally similar across the Great Lakes region, there is a unique climatic feature that generates vastly different quantities of snow across relatively small geographic areas. Lake-effect snow occurs when cold air moves over unfrozen lakes, picks up moisture, and deposits that moisture as snow on the downwind side of the lakes (Notaro et al., 2015; Burnett et al., 2003; Ellis and Johnson, 2004). Lake-effect snow leads to long-run seasonal snowfall totals that are remarkably different for neighboring regions, and generates large fluctuations in year-to-year snow totals. For example, Erie County, NY (which contains Buffalo) receives nearly twice as much snow as Cuyahoga County, OH (which contains Cleveland).⁵ These two cities are fewer than 200 miles apart, are both situated along Lake Erie's shoreline, share similar socioeconomic characteristics, have common industrial roots, and possess similar climates aside from differences in snow. Comparing these regions can isolate variation in snow from confounding climate and socioeconomic characteristics. LES also generates differential temporal variation in snowfall due to year-to-year changes in lake-ice cover. As lakes freeze, the intensity of LES diminishes, which implies that colder

winters result in less relative snowfall in lake-effect regions (i.e., winter temperatures and snow can move in opposite directions in lake-effect regions).

The lake-effect phenomenon provides a credible identification strategy for isolating the effect of snow on migration. By comparing migration responses across counties that regularly experience lake-effect snow (hereafter, “LES counties”) versus those that do not (“non-LES counties”), I can separate the impact of snow from other winter weather conditions. This comparison exploits two features of lake-effect snow. First, lake-effect and non-lake-effect counties within the same state experience dramatically different snowfall but nearly identical temperatures, precipitation patterns, and economic shocks, allowing me to isolate snow from confounders more convincingly than controlling for weather variables in a regression. Second, the comparison tests whether typical exposure matters for behavioral responses: if residents adapt to heavy snow through infrastructure, expectations, or selection, migration effects should be attenuated in lake-effect regions despite their higher snow levels.

Fig. 1 illustrates the geography of LES. The top right panel shows mean annual snow-water equivalent (SWE), described in more detail in the subsequent section, for the sample period 1970–2017. Darker areas indicate more annual snow. Michigan's upper peninsula, north-western Michigan, western New York, and upstate New York are the regions that experience the largest average annual snowfall. The bottom four panels display the spatial distribution of other meteorological variables. While precipitation is positively correlated with snow (which is

⁴ Source: <https://coast.noaa.gov/states/fast-facts/great-lakes.html> (Last accessed: January 28, 2025).

⁵ The 1970–2017 mean snow-water equivalent is 566 mm in Erie County, NY, and 305 mm in Cuyahoga County, OH.

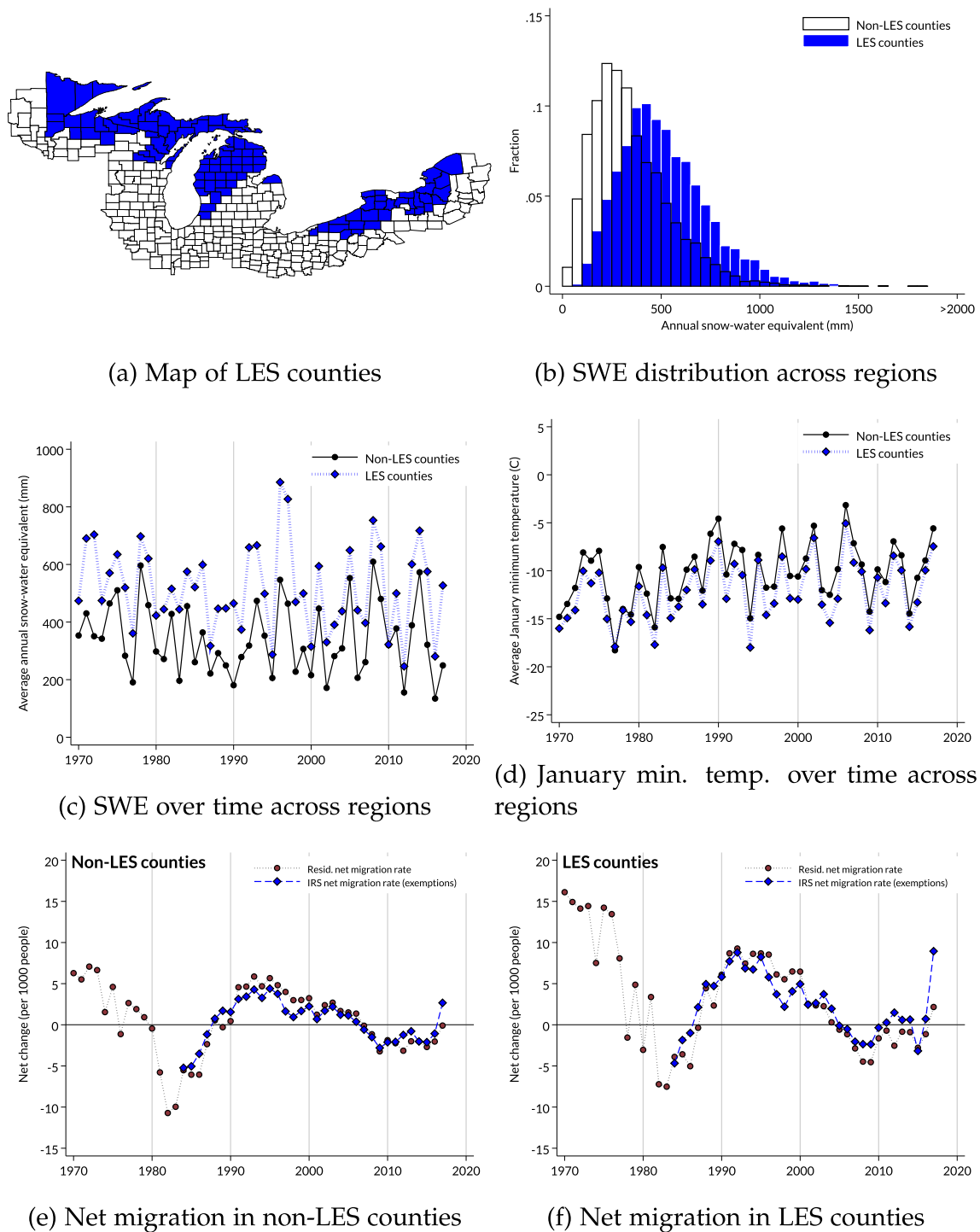


Fig. 2. Lake-effect snow regions, weather variables, and net migration outcomes. *Notes:* **Panel (a)** displays the assignment of counties to lake-effect snow (LES) and non-LES regions. LES counties are defined as those with centroids within 100 km of a Great Lakes shoreline and mean annual SWE above the median of all counties within 100 km of a Great Lake. **Panel (b)** shows the distribution of average annual SWE separately for LES and non-LES counties over 1970–2017. **Panels (c)** and **(d)** show population-weighted averages of annual SWE and January minimum temperature, respectively, for LES and non-LES counties over time. **Panels (e)** and **(f)** show annual averages of residual net migration and IRS net migration rates (based on exemptions) for non-LES and LES counties, respectively. Residual net migration is constructed following the forward residual method described in [Section 3.2](#). IRS net migration data are available from 1984 to 2017. All weather data are from TerraClimate ([Abatzoglou et al., 2018](#)).

unsurprising since precipitation must precede land-based snow measurements like SWE), the relationship is imperfect—high-snow regions like western New York and low-snow regions like southern Illinois receive similar total precipitation. Minimum temperatures generally decrease

with latitude, though Great Lakes shorelines exhibit moderating effects due to water temperature. Solar radiation varies primarily with latitude, showing smooth geographic gradients. Wind speed exhibits spatial heterogeneity that does not closely track snowfall patterns (e.g., high

winds in Cook County, IL, occur in a non-lake-effect region). Table A.1 further demonstrates that while LES and non-LES counties differ substantially in snowfall, other climate variables show more modest differences, isolating variation in snow effects across geographies.⁶

In the empirics, I categorize counties into discrete lake-effect regions, however there is no standard way to do this. Generally, LES is deposited within 80–100 km of a Great Lake shoreline, but can travel up to 150 km (Dewey, 1979; Norton and Bolsenga, 1993; Kunkel et al., 2002; Notaro et al., 2015). To assign counties to LES regions, I use the following rules: First, the county's centroid must fall within 100 km of a Great Lakes shoreline, approximating the distance that LES typically travels. Second, the county's mean annual SWE for 1970–2017 must be greater than the median of all counties within 100 km of a Great Lake, which approximates the average amount of regional snowfall that would be experienced in the absence of lake-effect snow (Kunkel et al., 2002).⁷

The top-left panel in Fig. 2 illustrates which counties are assigned to lake-effect regions. Comparing this figure to the long-run average of SWE in Fig. 1 suggests that the assignment effectively identifies counties that receive larger average snow on the southeastern side of a Great Lake. The top-right panel in Fig. 2 shows the distribution of SWE across regions. The bottom panels of Fig. 2 emphasize the temporal dynamics that reveal how comparison of LES regions to non-LES regions can isolate the effect of snow from other meteorological conditions. On the bottom left, average annual SWE is always greater in LES regions and often twice as large as SWE in non-LES regions. However, the bottom right panel shows that LES and non-LES regions have largely similar minimum January temperatures. LES regions are 1–2 degrees C colder on average, primarily due to LES regions being located further north, although the differences in temperature are small relative to the differences in annual snow across regions.

3. Data

The primary data sources include historical weather data from climate model outputs, as well as two types of migration data: one measured directly, and the other inferred indirectly from population and mortality records.

3.1. Weather data

I use historical weather information from the TerraClimate dataset (Abatzoglou et al., 2018), which includes monthly observations at a 4 km resolution from 1958 to 2017. I gather monthly average maximum and minimum temperatures, total precipitation, mean solar radiation, mean wind speed, and total SWE. To capture a continuous measure of each winter season, I define a “weather year” beginning on November 1 of the previous calendar year through October 31 of the current calendar year. For each month, I average observations for grid cells within a county weighted by block-level population from the 2000 Census. SWE is a measure of the amount of water contained in snow (e.g., if 1000 mm of snow has a 20% water density, then SWE is 200 mm). SWE serves as a proxy for experienced snowfall that is modeled consistently over space and long time-periods, whereas snow measurements at weather stations are inconsistent over time and contain many missing values, which pose challenges for empirical identification (Auffhammer et al., 2013).⁸ I limit the primary sample to counties with centroids that fall within 200 km of a Great Lake shoreline.⁹

⁶ In Figure A.1, I include annual maps for key weather variables from 1990 through 1999 to illustrate temporal variation in snow.

⁷ In the appendix, I show that results are robust to alternative distance bandwidths.

⁸ In the appendix, I show that the TerraClimate SWE data captures similar temporal variation to daily weather station observations of snowfall and snow depth obtained from the Global Historical Climatology Network.

⁹ In the appendix, I show that results are robust to alternative distance bandwidths, ranging from 100 km to 500 km.

Average statistics on weather variables are presented in Table A.1 across regions. The table shows that lake-effect counties have substantially larger SWE (around 175 mm per year), roughly 2 °C lower minimum and maximum temperatures, slightly lower solar radiation levels, and slightly smaller (~47 mm per year) precipitation levels than non-lake-effect counties. Wind speed is statistically similar across regions. Aside from SWE, the differences in climate variables are not large and are generally explained by the fact that lake-effect counties are located at higher latitudes.

3.2. Residual net migration

Migration data come from multiple sources. I first create a novel dataset of annual migration using the “residual” method following Passel et al. (2004); Feng et al. (2010, 2012); Chen et al. (2022); Baylis et al. (2025), among others. However, prior applications have calculated residual migration at a coarser temporal scale. For each county-year, “residual net migration” (M_{cy}) is given by:

$$M_{cy} = \text{Pop}_{cy} - \underbrace{[\text{Pop}_{cy-1} + \text{Births}_{cy} - \text{Deaths}_{cy}]}_{\text{Expected population}}. \quad (1)$$

The difference between this year's population and last year's population, net of natural rates of change, defines residual net migration.¹⁰ All measures of net migration are converted into rates (m_{cy}) by dividing by the previous year's population in 1000s. I obtain data for each component on the right-hand side of Eq. (1) to construct an annual measure of residual net migration.

I use annual county-level population (Pop_{cy}) from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program data, from 1969 through 2017. SEER population data is used to calculate official annual mortality and cancer rates by age, sex, and race. I use the “less than one-year old” population to proxy annual births. SEER population data is constructed from Census products and imputed between decennial Census population counts.¹¹ The imputation process follows the Census Bureau's Population Estimates Program methodology for calculating intercensal population estimates. As a result, these data are measured with error, but provide the some of the best available annual population data at the county level.

I obtain annual mortality data from the National Center for Health Statistics (NCHS). NCHS provides publicly available mortality data from 1968 to 1988 by age-group for all counties, after which public data is only available for large counties. For 1989–2017, I obtain restricted-use data from NCHS, which provides mortality data for all counties by age group. I use reported county of residence at time of death to calculate migration rates.

3.3. IRS net migration data

The second source of migration data is annual, county-level migration derived from tax returns. The Internal Revenue Service (IRS) publishes migration flows that are identified by year-to-year changes in the residence of tax filers. IRS data are available from 1991 through 2017. I supplement the IRS data with historical IRS migration data from the Inter-university Consortium for Political and Social Research provided for 1984–1990.¹² For each county, the IRS data set reports the quantity of in-migrants, out-migrants, and non-migrants. I construct an “IRS net migration rate” as the difference between in-migrants and out-migrants divided by the sum of non-migrants and out-migrants per 1000.¹³ I use the number of exemptions on each tax return, potentially

¹⁰ In Appendix A.3, I describe an analogous approach to measure age-specific net migration.

¹¹ See more details here: <https://seer.cancer.gov/popdata/modifications.html>.

¹² These data are accessible here: <https://doi.org/10.3886/ICPSR02937.v1>.

¹³ This approach is adopted from the IRS's approach, reported here: <https://www.irs.gov/pub/irs-soi/migrationdatacomparisons.pdf>.

representing children or non-working spouses, to better approximate the quantity of *people* in calculating migration rates. Because the IRS data are based on tax filings, poor and very rich residents, young and elderly residents, and undocumented residents are likely underrepresented.

The IRS data allow for construction of in- and out-migration rates, as well as migration flows by origin and destination region.¹⁴ I use these data to explore whether the net migration response to snow is driven by changes in in- or out-migration and whether migrants are moving within-state or out-of-state.

Both measures of net migration are presented in panels (e) and (f) in Fig. 2. Despite coming from different sources, the two measures of net migration are remarkably similar for overlapping time periods.¹⁵ Net migration declines between 1970 and 1982, after which it increases until the mid-1990s, and then gradually declines again until the mid-2010s. The trends in net migration are similar across LES and non-LES regions, although the magnitude of changes in the net migration rate is larger in LES regions.

4. Empirical strategy

The goal of the empirical analysis is to estimate whether snow influences net migration. Let $m_{c,sy}$ be the net migration rate (per 1000 people) in county c , in state s , in year y , which is either the residual or IRS net migration rate. The primary estimating equation is:

$$m_{c,sy} = \beta_1 \cdot 1[\text{SWE}_{cy-k} > 110\% \times \overline{\text{SWE}_{cy}}] + \mathbf{W}'_{cy-k} \boldsymbol{\theta} + \alpha_c + \lambda_{sy} + \varepsilon_{c,sy} \quad (2)$$

where $1[\text{SWE}_{cy-k} > 110\% \times \overline{\text{SWE}_{cy}}]$ is an indicator for whether annual SWE in county c in year $y - k$ (for $k = 0, 1, 2$ lags) exceeds 110% of the county's 10-year rolling mean, $\overline{\text{SWE}_{cy}}$.¹⁶ $\mathbf{W}_{cy-k}$ is a vector of weather controls in county c in year $y - k$, which includes monthly averages of minimum and maximum temperatures, solar radiation, wind speed, and non-winter precipitation, all lagged by k periods to match the SWE lag.¹⁷ α_c is a county fixed effect and λ_{sy} is a state-by-year fixed effect. Standard errors are clustered at the county level to account for serial correlation within counties over time.

The 10-year rolling mean ($\overline{\text{SWE}_{cy}}$) is calculated as mean SWE in county c over the 10 years preceding year y . The 110% threshold defines "above-normal" snowfall: years when a county receives at least 10% more snow than its recent average. This binary specification provides an easily interpretable measure of unusually snowy winters while allowing for potential nonlinearities in the migration response to snow.

I estimate regressions for k lags ranging from zero to two years. Because migration rates are defined as changes from the previous year, contemporaneous weather is unlikely to influence migration. The primary focus is on the one-year lag ($k = 1$), with $k = 0$ serving as a placebo check and $k = 2$ revealing longer-run dynamics.

Weather controls enter as monthly levels rather than deviations because: (1) they exhibit meaningful seasonal variation requiring within-year controls, (2) their year-to-year variation is considerably smaller than snow, and (3) they serve to control for absolute contemporaneous conditions. In robustness checks presented in the appendix, alternative specifications including annual averages, annual deviations from longer-run means, and inclusion of winter precipitation generate similar results.

¹⁴ However, the IRS suppresses flows when fewer than 10 tax returns move between a specific origin–destination pair and aggregates these returns into broader categories.

¹⁵ I explore the similarity of these measures in more detail in Appendix A.11.

¹⁶ TerraClimate data begins in 1958, so I cannot construct longer-run averages for earlier periods. Ten years was chosen due to data limitations and as a reasonable proxy for recently experienced weather.

¹⁷ Winter precipitation is excluded due to its mechanical relationship with SWE: snow must fall as precipitation before becoming SWE.

Consider the specification in Eq. (2) with $k = 1$ for illustration. This equation estimates the effect of above-normal SWE last year on net migration rates, conditional on weather controls and fixed effects. The inclusion of county fixed effects (α_c) and state-by-year fixed effects (λ_{sy}) allows β_1 to be interpreted as the effect of an above-normal snow year relative to normal or below-normal years within the same county, controlling for common state-level temporal shocks (e.g., policy changes, regional economic conditions). Because the sample is limited to counties within 200 km of a Great Lake, state-by-year fixed effects capture sub-state regional variation.

I estimate Eq. (2) for the full sample and separately for LES and non-LES counties to assess whether migration responses differ across regions with varying snow exposure. This split-sample approach is more flexible than pooled interaction specifications because it allows weather controls and fixed effects to vary by LES designation, mitigating potential omitted-variable bias (Feigenberg et al., 2023).¹⁸

4.1. Nonlinearities, directional flows, dynamics, and heterogeneity

I explore four extensions to the primary specification, each estimated separately for LES and non-LES counties to assess differential responses across regions with varying snow exposure. First, to examine nonlinearities in the snow–migration relationship, I replace the binary indicator in Eq. (2) with binned percentage deviations from $\overline{\text{SWE}_{cy}}$: >50% above/below normal, 30–50% above/below normal, and 10–30% above/below normal, with the excluded bin capturing deviations within $\pm 10\%$. Second, using IRS data, I decompose net migration into in-migration and out-migration rates, and further separate migrants by origin and destination (in-state vs. out-of-state) to distinguish whether snow deters potential arrivals or accelerates departures. Third, I estimate a distributed lag model by replacing the single indicator in Eq. (2) with $\sum_{k=-2}^5 \beta_k \cdot 1[\text{SWE}_{cy-k} > 110\% \times \overline{\text{SWE}_{cy}}]$, which includes five lags ($k = 1, \dots, 5$), contemporaneous ($k = 0$), and two leads ($k = -1, -2$) relative to year y ; weather controls are included for $k = 0, 1, 2$. Fourth, using residual net migration data, I estimate age-specific migration responses for 10-year cohorts to identify which demographic groups are most responsive to snow.

5. Results

5.1. Main effects of above-normal snow on net migration rates

I present the baseline results in Table 1. Panel A shows the results for residual net migration, while panel B shows the results for IRS net migration. Within each panel, I present coefficients on above-normal snowfall for 0, 1, and 2, lags for the full sample and separated by lake-effect and non-lake-effect counties. For both outcomes, there is no significant effect for contemporaneous snow ($k = 0$), which is consistent with the fact that migration rates are defined as changes from the previous year. For lagged snow ($k = 1$), there is a significant negative effect of above-normal snow on net migration for the full sample, which is driven by a larger negative effect in lake-effect counties. The magnitude ranges from -1.6 to -3.1 people per 1000, for each outcome. The effect in non-lake-effect counties is small and statistically zero. The results for $k = 2$ are similar but slightly attenuated relative to $k = 1$.

The concentration of effects in lake-effect counties, despite non-lake-effect counties also experiencing substantial year-to-year snow variation, rules out insufficient variation or omitted variables correlated with winter severity as explanations. Instead, heavy snow affects migration only when it is a recurring feature of local climate, consistent with belief updating about future conditions or threshold effects from repeated exposure. Collectively, these results suggest that higher-than-normal SWE in the preceding two years reduces net migration rates, and

¹⁸ Results using LES interactions in a pooled specification are included in the appendix.

Table 1

The effect of above-normal snow events on net migration outcomes.

Panel A: Residual net migration, per 1000 population									
	Lag: 0 years			Lag: 1 year			Lag: 2 years		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Full	LES = 0	LES = 1	Full	LES = 0	LES = 1	Full	LES = 0	LES = 1
$1[\text{SWE}_{cy} > 110\% \times \overline{\text{SWE}}_{cy}]$	0.17 (0.37)	0.29 (0.32)	0.30 (0.97)						
$1[\text{SWE}_{cy-1} > 110\% \times \overline{\text{SWE}}_{cy}]$				−1.17*** (0.37)	0.05 (0.32)	−3.09*** (0.90)			
$1[\text{SWE}_{cy-2} > 110\% \times \overline{\text{SWE}}_{cy}]$							−0.31 (0.47)	0.60 (0.43)	−2.67** (1.12)
Obs.	17,376	12,480	4896	17,014	12,220	4794	16,652	11,960	4692
Dep. var. mean	1.25	0.49	3.19	1.08	0.36	2.91	0.93	0.25	2.65
Weather controls	Y	Y	Y	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y
State × Year FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y
Panel B: IRS net migration (exemptions), per 1000 population									
	Lag: 0 years			Lag: 1 year			Lag: 2 years		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Full	LES = 0	LES = 1	Full	LES = 0	LES = 1	Full	LES = 0	LES = 1
$1[\text{SWE}_{cy} > 110\% \times \overline{\text{SWE}}_{cy}]$	0.43 (0.38)	0.01 (0.29)	1.80 (1.29)						
$1[\text{SWE}_{cy-1} > 110\% \times \overline{\text{SWE}}_{cy}]$				−0.50** (0.25)	0.04 (0.26)	−1.61** (0.73)			
$1[\text{SWE}_{cy-2} > 110\% \times \overline{\text{SWE}}_{cy}]$							0.01 (0.22)	0.20 (0.24)	−0.78 (0.55)
Obs.	12,305	8840	3465	12,305	8840	3465	12,305	8840	3465
Dep. var. mean	0.92	0.34	2.40	0.92	0.34	2.40	0.92	0.34	2.40
Weather controls	Y	Y	Y	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y
State × Year FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Dependent variables are net migration rates per 1000 population at the county-year level. Coefficients presented represent “above-normal snow” which is defined as $1[\text{SWE}_{cy-k} > 110\% \times \overline{\text{SWE}}_{cy}]$ where SWE_{cy-k} is snow-water equivalent in county c in year $y - k$, $\overline{\text{SWE}}_{cy}$ is the 10-year moving average of SWE in county c preceding year y , and k indicates the lag structure (0, 1, or 2 years). Columns are organized by lag, with estimates for full sample (“Full”), non-lake-effect counties (“LES = 0”), and lake-effect counties (“LES = 1”). Weather controls include (lagged) monthly averages of minimum and maximum temperatures, solar radiation, wind speed, and non-winter precipitation. Standard errors are clustered at the county level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

this effect is primarily driven by responses in lake-effect counties, which receive more snow on average.

5.2. Nonlinear effects of snow deviations on net migration

Next, I explore potential nonlinearities in the effect of snow deviations on net migration. I present these results graphically in Fig. 3 for residual net migration (panel (a)) and IRS net migration (panel (b)). For both outcomes, I plot the coefficients on binned SWE deviations for one-year lags. Each plot shows estimates from separate regressions: full sample (black circles), lake-effect counties (blue triangles), and non-lake-effect counties (gray squares). The omitted category is the “normal” range of SWE deviations (within 10 percentage points of $\overline{\text{SWE}}_{cy}$).

The nonlinear effects confirm the results of Table 1. Lower than normal snowfall results in no statistical change or small increases in net migration rates for both lake-effect and non-lake-effect counties and for both outcomes. In contrast, higher than normal snow results in decreases in net migration rates for lake-effect counties, with small and statistically insignificant changes for non-lake-effect counties. Notably, the bin-specific effects are relatively flat across above-normal snow bins: even modest anomalies (10–30% above normal) generate effects similar in magnitude to more extreme anomalies (>50% above normal), with overlapping confidence intervals across bins. This pattern is consistent with a discrete rather than continuous migration response—any above-normal snow year appears sufficient to deter migration, regardless of its severity. One interpretation is that would-be migrants respond to the occurrence of an above-normal snow year as a signal about local

climate rather than to its intensity. This motivates the use of a binary above-normal indicator as the primary specification and for the following explorations of heterogeneity. Overall, these results suggest that the migration response to snow is driven by above-normal snow in lake-effect regions, which receive more snow on average, and that there is no meaningful response to above-normal snow in non-lake-effect regions.

5.3. Directional and regional migration flows

In Fig. 4(a), I decompose the IRS net migration response into in-migration and out-migration components and explore heterogeneity by geographic origin and destination. The top panels show that above-normal snow in lake-effect counties has no significant effect on out-migration rates but reduces in-migration by approximately 1.5 people per 1000. Naturally, the effect on net migration is a combination of the in- and out-migration effects, which delivers an overall reduction in net migration in response to snow for lake-effect counties. In contrast, non-lake-effect counties exhibit smaller and marginally significant positive effects on in- and out-migration rates. These two effects net out to no statistical change in net migration. This pattern suggests that snow primarily deters potential in-migrants to lake-effect regions rather than prompting current residents to leave.

The coefficient plots in Fig. 4(a), also decompose the directional flows by region (in-state vs. out-of-state). These results are more tenuous partly due to censoring of the IRS data for small flows, which attenuates the coefficients. Nonetheless, the results suggest that the reduction in in-migration to lake-effect counties is driven by declines in both in-state

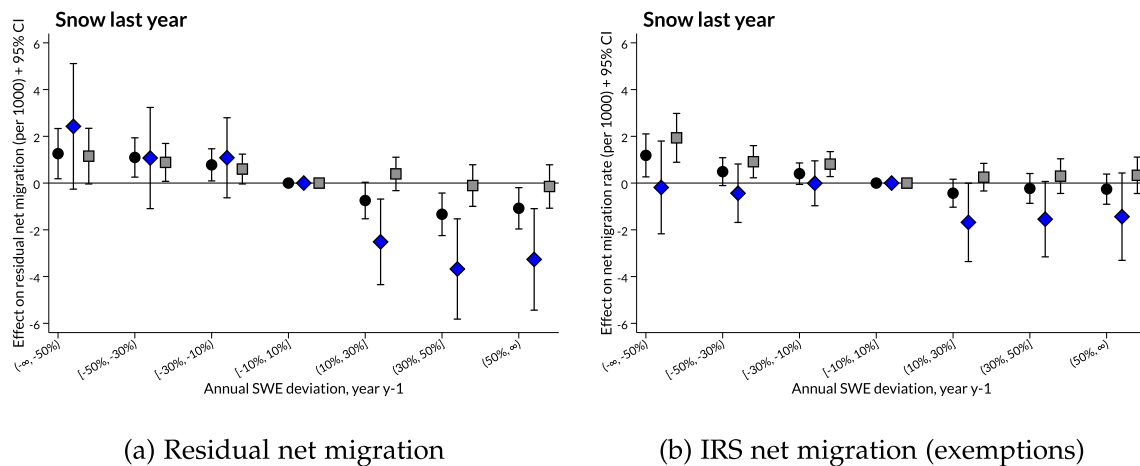


Fig. 3. Results for binned snow-water equivalent (SWE) deviations on net migration outcomes. *Notes:* Each panel presents coefficients for a dummy variable indicating bins of percentage deviations from 10-year mean SWE. The omitted category is the “normal” range of SWE deviations, which is defined as being within 10 percentage points of 10-year mean SWE. Regressions include weather controls, state-by-year fixed effects, and county fixed effects. The estimates are presented with 95% confidence intervals, clustered at the county level. Separate regressions are estimated for each outcome variable and time lag.

and out-of-state migration, with a slightly larger effect for in-state flows. For non-LES counties, there is a slight increase in within-state in- and out-migration flows in response to above normal snow, which cancel out on net.

5.4. Dynamics of snow on net migration

In Figs. 4(b) and 4(c), I present results from distributed lag models for both migration outcomes. Each panel shows coefficients on two leads, contemporaneous, and five lags of above-normal snow, along with weather controls, county fixed effects, and state-by-year fixed effects. For non-LES counties, all coefficients are close to zero and generally insignificant for both outcomes. For LES counties, however, the coefficients leading up to and including the above-normal snow event are statistically zero, suggesting no pre-trends in migration prior to the abnormally snowy winters. In the year after above-normal snow, both outcomes show a significant decrease in net migration rates, which is consistent with the main results. The effect for residual net migration persists for two years after the snow shock, with some weak evidence of longer-run effects, while the effect for IRS net migration is only significant for one year after. The magnitude of this effect is a 2–3 person per 1000 reduction in net migration rates, which is comparable across outcomes and consistent with the main results. Overall, these results suggest that the effect of short-run variation in snow on migration is transitory, which provides further evidence that the primary effects are driven by behavioral responses to weather shocks rather than spurious correlations with longer-run changes in climate or other confounding factors.

5.5. Heterogeneity by age

Lastly, in Fig. 4(d), I explore the effects of above-normal snow on age-specific residual migration rates.¹⁹ The results show significant negative effects of lagged above-normal SWE on residual net migration for all 10-year age cohorts up to the 60–69 year cohort in lake-effect regions. Cohorts aged 70 and older have insignificant effects. The largest effects are for 20–29 year-olds (reductions of approximately 4 people per 1000), which decline somewhat as cohorts grow older. Despite minor differences, confidence intervals overlap across cohorts, suggesting that the effect of snow on migration is not meaningfully different across

younger and middle age groups. Interestingly, the results suggest that the effect of snow is modest for cohorts squarely within the retirement age group (i.e., 60–69) and nonexistent for older adults (i.e., 70 and above). For non-lake-effect counties, the pattern is starkly different. All effects are close to zero, and there are small decreases for the 50–59 and 60–69 age groups. This lack of heterogeneity and small magnitudes corroborates the main results that snow does not seem to have a meaningful effect on net migration for non-lake-effect counties.

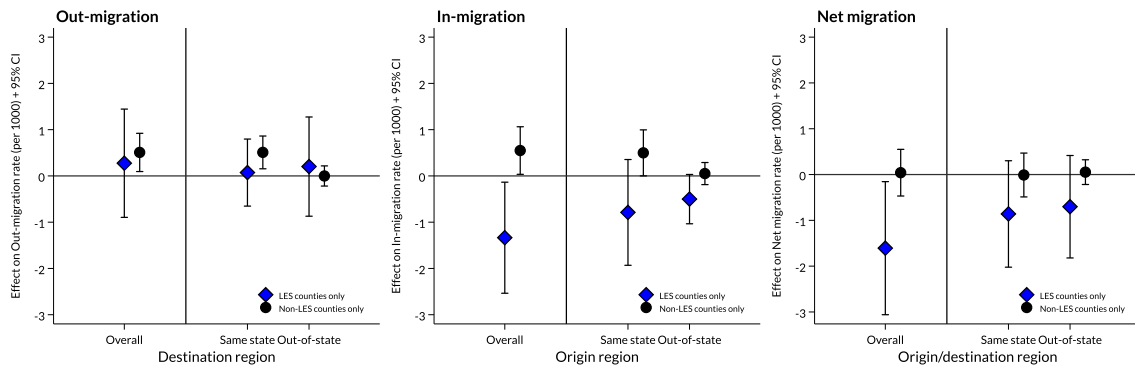
5.6. Additional results and sensitivity checks

I implement a variety of tests to probe the sensitivity of the main results, which are discussed in the appendix and briefly summarized here. The primary results are robust to alternative weather controls, fixed effects specifications, alternative definitions of snow, and alternative comparisons of snow effects across lake-effect regions (Appendix A.4, A.5, A.6). The results are also robust to varying the geographic sample restrictions and lake-effect county assignments (Appendix A.7). I also explore whether migration responds to longer-run changes in SWE following Bento et al. (2023) (Appendix A.6). Short-run effects are about 19–38% the magnitude of a one-standard deviation (~44%) increase in the 10-year moving average of SWE in lake-effect counties, suggesting that annual deviations in snow generate substantial migration responses relative to longer-run changes. I also find that the effect of snow on migration is more pronounced in earlier decades (Appendix A.10). Further, there is no meaningful evidence of a direct effect of snow on mortality (all-cause or traffic fatalities) or birth rates, which supports the validity of the residual migration measure (Appendix A.13). Finally, TerraClimate SWE compares favorably to weather station data, though station-based estimates are attenuated due to data quality issues (Appendix A.14).

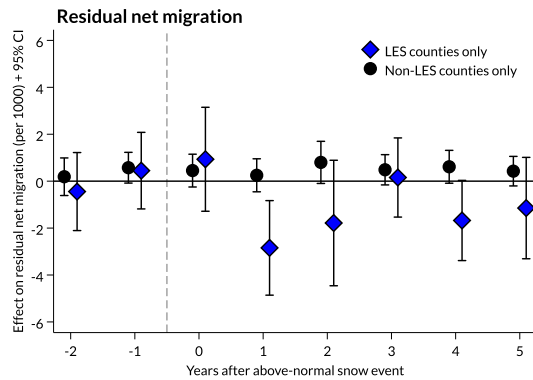
6. Discussion and conclusion

This paper provides new evidence on the effect of short-run variation in weather on migration. Abnormally large amounts of snow generate losses in net population in the following years. This effect is concentrated in regions that regularly receive large amounts of snow, and is driven by reductions in in-migration. This result suggests that short-run variation in weather does affect revealed preference values of local climate disamenities, which in turn suggests that cross-sectional analyses of households' willingness-to-pay for climate amenities may not capture the full effect of climate on migration. Further, these findings imply that

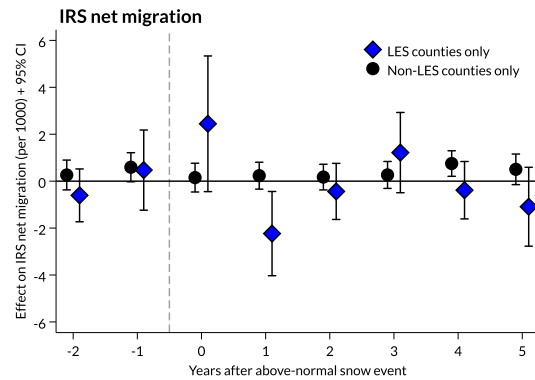
¹⁹ The construction of age-specific migration rates is described in Appendix A.3.



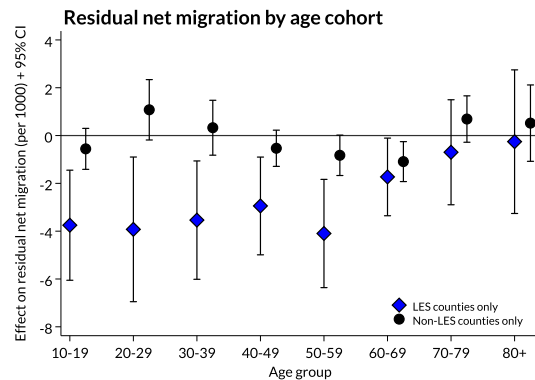
(a) In-/out-migration and origin/destination heterogeneity (IRS net migration)



(b) Distributed lag (Resid. net migration)



(c) Distributed lag (IRS net migration)



(d) Age heterogeneity (Resid. net migration)

Fig. 4. Heterogeneity and dynamics of snow on net migration. *Notes:* **Panel (a)** presents coefficients for an indicator of above-normal snow in the previous year on in-migration (left), out-migration (center), and net migration (right) using IRS tax return data. Results are also presented for in-state vs. out-of-state flows. Each regression includes county fixed effects, state-by-year fixed effects, and lagged monthly weather controls (minimum and maximum temperatures, solar radiation, wind speed, and non-winter precipitation). Separate regressions are estimated for non-LES counties (black circles) and LES counties (blue diamonds). 95% confidence intervals are presented, clustered at the county level. **Panels (b) and (c)** present distributed lag regressions where net migration rates are regressed on two leads, contemporaneous, and five lags (i.e., $k = -2, -1, \dots, 5$) of an indicator for above-normal snow ($1[\text{SWE}_{cy-k} > 110\% \times \overline{\text{SWE}_{cy}}]$), along with monthly weather controls (contemporaneous plus two lags), county fixed effects, and state-by-year fixed effects. Weather controls include monthly averages of minimum and maximum temperatures, solar radiation, wind speed, and non-winter precipitation. Separate regressions are estimated for non-LES counties (black circles) and LES counties (blue diamonds). 95% confidence intervals are presented, clustered at the county level. **Panel (d)** presents coefficients on $1[\text{SWE}_{cy-1} > 110\% \times \overline{\text{SWE}_{cy}}]$ for residual net migration by age cohort. Separate regressions are estimated for LES and non-LES counties for each 10-year age cohort. 95% confidence intervals are presented, clustered at the county level.

studies focused only on temperature and precipitation omit an important causal factor of weather on migration outcomes.

How big is the effect of snow on migration? In lake-effect counties, an above-normal snow year reduces net migration by 1.5–3 people per 1000 in the following year. The average lake-effect county had 60,000 residents in 1970, which translates to a reduction of 90–180 people after

an above-normal snow year.²⁰ An alternative way to frame the magnitude is to rescale the primary coefficients into changes as a percent

²⁰ Population increased over the sample period for the average county, so the conversions presented here are conservative.

of population by multiplying by 0.1 (i.e., undoing the “per 1000” scaling and multiplying by 100 to translate into percentage points). The average annual effect of above-normal snow in LES counties leads to a 0.15–0.3 percentage-point decrease in population. A recent study of annual county-level migration across the continental US over similar time periods found that the effect of 100 additional cooling-degree days per year reduced average net migration rates by 0.03 percentage points (Baylis et al., 2025). Taken directly, this comparison suggests that an above-normal snow year generates 5–10 times the migration loss that would be produced by increasing annual temperatures by 2 deg C for 50 days of the year. Or, based on long-difference estimates from the same paper, an above-normal snow year generates approximately 40–75% of the migration-response to increasing longer-run mean temperatures by 100 additional cooling-degree days per year.²¹ This comparison suggests that the magnitude of the effect of snow on migration is substantial and that regularly experienced annual deviations in snow have a more pronounced effect than large changes in annual temperatures.

The effect of snow on migration is driven by regions that regularly receive large amounts of snow, which suggests that short-run variation in weather interacts with longer-run local climates. Conventional wisdom might suggest that the marginal effect of snow would be *smaller* in regions that regularly receive heavy snow due to historical adaptation, although the results suggest the opposite. This puzzle may be explained by how (potential) residents update their beliefs about the future distribution of weather events. If (potential) residents in non-lake-effect regions view an unusually snowy winter as an anomalous event, then it is unlikely to affect migration patterns. However, unusually large amounts of snow in a lake-effect region may be indicative of the likelihood of these events occurring in the future. Or, if residents repeatedly experience snowy winters then an additional bad winter may serve as “the straw that broke the camel’s back.” However, given the evidence of the “deterrent” effect of snow on in-migration, this latter explanation is a less plausible mechanism.

This research is limited by several factors that provide avenues for future work. First, I have not directly accounted for how residents trade off climate amenities for local wages and rents, or how those tradeoffs have evolved over time. Low wages or high unemployment may interact with snow, potentially amplifying its negative effect on migration. Future work disentangling the value of winter weather from local economic conditions would provide a more complete picture of residential location choice. Second, it is unclear how these results generalize to other regions, though focusing on the Northeast and Midwest is valuable given the relative population decline in this region, typical exposure to winter weather, and status as a potential “climate haven.” Lastly, the results suggest that snow may have a transitory effect on migration. Future work could further explore the dynamics of adaptation to short-run weather shocks and how people update beliefs about future climate conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.jpubeco.2026.105637.

Data availability

Replication data and code are available at <https://doi.org/10.5281/zenodo.19237866>.

References

- Abatzoglou, J.T., Dobrowski, S.Z., Parks, S.A., Hegewisch, K.C., 2018. TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958–2015. *Sci. Data* 5 (1), 1–12.
- Albouy, D., Graf, W., Kellogg, R., Wolff, H., 2016. Climate amenities, climate change, and American quality of life. *J. Assoc. Environ. Resour. Econ.* 3 (1), 205–246.
- Auffhammer, M., Hsiang, S.M., Schlenker, W., Sobel, A., 2013. Using weather data and climate model output in economic analyses of climate change. *Rev. Environ. Econ. Policy* 7 (2), 181–198.
- Bakkensen, L.A., Lala, M., 2020. Sorting over flood risk and implications for policy reform. *J. Environ. Econ. Manag.* 104, 102362.
- Barreca, A., Clay, K., Deschenes, O., Greenstone, M., Shapiro, J.S., 2016. Adapting to climate change: the remarkable decline in the US temperature-mortality relationship over the twentieth century. *J. Polit. Econ.* 124 (1), 105–159.
- Baylis, P., Bharadwaj, P., Mullins, J.T., Obradovich, N., 2025. Climate and migration in the United States. *J. Publ. Econ.* 249, 105446.
- Bento, A.M., Miller, N., Mookerjee, M., Severini, E., 2023. A unifying approach to measuring climate change impacts and adaptation. *J. Environ. Econ. Manag.* 102843.
- Bohra-Mishra, P., Oppenheimer, M., Hsiang, S.M., 2014. Nonlinear permanent migration response to climatic variations but minimal response to disasters. *Proc. Natl. Acad. Sci.* 111 (27), 9780–9785.
- Boustian, L.P., Kahn, M.E., Rhode, P.W., Yanguas, M.L., 2020. The effect of natural disasters on economic activity in US counties: a century of data. *J. Urban Econ.* 118, 103257.
- Burnett, A.W., Kirby, M.E., Mullins, H.T., Patterson, W.P., 2003. Increasing great lake-effect snowfall during the twentieth century: a regional response to global warming? *J. Clim.* 16 (21), 3535–3542.
- Cattaneo, C., Beine, M., Fröhlich, C.J., Kniveton, D., Martinez-Zarzoso, I., Mastrorillo, M., Millock, K., Piguat, E., Schraven, B., 2019. Human migration in the era of climate change. *Rev. Environ. Econ. Policy* 13 (2), 189–206.
- Chen, S., Oliva, P., Zhang, P., 2022. The effect of AIR pollution on migration: evidence from China. *J. Dev. Econ.* 156, 102833.
- Cragg, M., Kahn, M., 1997. New estimates of climate demand: evidence from location choice. *J. Urban Econ.* 42 (2), 261–284.
- Deryugina, T., 2013. How do people update? The effects of local weather fluctuations on beliefs about global warming. *Clim. Change* 118, 397–416.
- Deryugina, T., Molitor, D., 2020. Does when you die depend on where you live? Evidence from Hurricane Katrina. *Am. Econ. Rev.* 110 (11), 3602–3633.
- Dewey, K.F., 1979. An objective forecast method developed for Lake Ontario induced snowfall systems. *J. Appl. Meteorol. Climatol.* 18 (6), 787–793.
- Ellis, A.W., Johnson, J.J., 2004. Hydroclimatic analysis of snowfall trends associated with the North American Great Lakes. *J. Hydrometeorol.* 5 (3), 471–486.
- Eyer, J., Dinterman, R., Miller, N., Rose, A., 2018. The effect of disasters on migration destinations: evidence from Hurricane Katrina. *Econ. Disasters Clim. Change* 2 (1), 91–106.
- Fan, Q., Fisher-Vanden, K., Klaiber, H.A., 2018. Climate change, migration, and regional economic impacts in the United States. *J. Assoc. Environ. Resour. Econ.* 5 (3), 643–671.
- Feigenberg, B., Ost, B., Qureshi, J.A., 2023. Omitted variable bias in interacted models: a cautionary tale. *Rev. Econ. Stat.* 1–47.
- Feng, S., Krueger, A.B., Oppenheimer, M., 2010. Linkages among climate change, crop yields and Mexico-US cross-border migration. *Proc. Natl. Acad. Sci.* 107 (32), 14257–14262.
- Feng, S., Oppenheimer, M., Schlenker, W., 2012. Climate Change, Crop Yields, and Internal Migration in the United States. Technical Report, National Bureau of Economic Research.
- Gibson, J., McKenzie, D., 2011. Eight questions about brain drain. *J. Econ. Perspect.* 25 (3), 107–128.
- Glaeser, E.L., Oct 2007. Can Buffalo ever come back? *The New York Sun*.
- Glaeser, E.L., Tobio, K., 2008. The rise of the Sunbelt. *South. Econ. J.* 74 (3), 609–643.
- Graves, P.E., 1980. Migration and climate. *J. Reg. Sci.* 20 (2), 227–237.
- Hauer, M.E., 2017. Migration induced by sea-level rise could reshape the US population landscape. *Nat. Clim. Change* 7 (5), 321–325.
- Hornbeck, R., 2012. The enduring impact of the American Dust Bowl: short-and long-run adjustments to environmental catastrophe. *Am. Econ. Rev.* 102 (4), 1477–1507.
- Hsiang, S., Kopp, R., Jina, A., Rising, J., Delgado, M., Mohan, S., Rasmussen, D.J., Muir-Wood, R., Wilson, P., Oppenheimer, M., Larsen, K., Houser, T., 2017. Estimating economic damage from climate change in the United States. *Science* 356 (6345), 1362–1369.
- IPCC, 2022. Climate Change 2022: Impacts, Adaptation, and Vulnerability. Vol. Cambridge University Press. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.
- Kahn, M.E., 2013. *Climatopolis: How Our Cities Will Thrive in the Hotter Future*. Basic Books (AZ). pp. 1–288. <https://www.hachettebookgroup.com/titles/matthew-e-kahn/climatopolis/9780465063833/?lens=basics-books>.
- Kunkel, K.E., Westcott, N.E., Kristovich, D.A.R., 2002. Assessment of potential effects of climate change on heavy lake-effect snowstorms near Lake Erie. *J. Great Lakes Res.* 28 (4), 521–536.

²¹ Baylis et al. (2025) estimate, using long-differences across 1983–1987 and 2014–2018, that 100 additional cooling degree days generate a –0.4 percentage point change in net migration rates.

- Leduc, S., Wilson, D.J., 2024. Snow belt to Sun Belt migration: end of an era? Federal Reserve Bank of San Francisco Working Paper. pp. 21.
- Mahajan, P., Yang, D., 2020. Taken by storm: hurricanes, migrant networks, and US immigration. *Am. Econ. J.: Appl. Econ.* 12 (2), 250–277.
- Molloy, R., Smith, C.L., Wozniak, A., 2011. Internal migration in the United States. *J. Econ. Perspect.* 25 (3), 173–196.
- Moore, F.C., Obradovich, N., Lehner, F., Baylis, P., 2019. Rapidly declining remarkability of temperature anomalies may obscure public perception of climate change. *Proc. Natl. Acad. Sci.* 116 (11), 4905–4910.
- Norton, D.C., Bolsenga, S.J., 1993. Spatiotemporal trends in lake effect and continental snowfall in the Laurentian Great Lakes, 1951–1980. *J. Clim.* 6 (10), 1943–1956.
- Notaro, M., Bennington, V., Vavrus, S., 2015. Dynamically downscaled projections of lake-effect snow in the Great Lakes Basin. *J. Clim.* 28 (4), 1661–1684.
- Obolensky, M., Tabellini, M., Taylor, C., 2024. Homeward Bound: How Migrants Seek Out Familiar Climates. Technical Report, National Bureau of Economic Research.
- Passey, J.S., Van Hook, J., Bean, F.D., 2004. Estimates of the Legal and Unauthorized Foreign-Born Population for the United States and Selected States, Based on Census 2000. US Bureau of the Census, Washington, DC.
- Roback, J., 1982. Wages, rents, and the quality of life. *J. Polit. Econ.* 90 (6), 1257–1278.
- Sheldon, T.L., Zhan, C., 2022. The impact of hurricanes and floods on domestic migration. *J. Environ. Econ. Manag.* 115, 102726.
- Sinha, P., Caulkins, M.L., Cropper, M.L., 2018. Household location decisions and the value of climate amenities. *J. Environ. Econ. Manag.* 92, 608–637.
- Sinha, P., Caulkins, M., Cropper, M., 2021. The value of climate amenities: a comparison of hedonic and discrete choice approaches. *J. Urban Econ.* 126, 103371.
- Timmins, C., 2007. If you cannot take the heat, get out of the cerrado...recovering the equilibrium amenity cost of nonmarginal climate change in Brazil. *J. Reg. Sci.* 47 (1), 1–25.
- Winkler, R.L., Rouleau, M.D., 2021. Amenities or disamenities? Estimating the impacts of extreme heat and wildfire on domestic US migration. *Popul. Environ.* 42, 622–648.
- Wrenn, D.H., 2024. The effect of natural disasters and extreme weather on household location choice and economic welfare. *J. Assoc. Environ. Resour. Econ.* 11 (5), 1101–1134.